

Origin of crack propagation in LiCoO₂ cathode

Wenguang Zhao,^{1, †} Zijian Li,^{2, †} Mingyang Li,^{3, †} Hengyu Ren,¹ Cong Lin,⁴ Haocong Yi,¹ Shengyu Wu,⁵ Xiaohu Wang,¹ Jun Wang,⁶ Bin Fei,² Muyu Zhang,⁷ Mingwei Wang,⁸ Ye Sun,⁸ Guojie Chen,¹ Zhefeng Chen,¹ Weixin Chen,⁹ Xiaoxu Zhao,⁸ Shunning Li,^{1,10, *} Dong Zhou,^{11, *} Qinghe Zhao,^{1,5, *} Feng Pan^{1, *}

¹ School of Advanced Materials, Peking University Shenzhen Graduate School, Shenzhen 518055, China.

² School of Fashion and Textiles, The Hong Kong Polytechnic University, Kowloon 999077, Hong Kong S.A.R., China.

³ Department of Mechanical and Energy Engineering, Southern University of Science and Technology, Shenzhen 518055, China.

⁴ Department of Applied Biology and Chemical Technology, The Hong Kong Polytechnic University, Kowloon, Hong Kong S.A.R., China.

⁵ College of Physics and Energy, Fujian Normal University, Fujian Provincial Key Laboratory of Quantum Manipulation and New Energy Materials, Fuzhou, 350117, China.

⁶ School of Innovation and Entrepreneurship, Southern University of Science and Technology, Shenzhen 518055, China.

⁷ Department of Mechanics and Aerospace Engineering, Southern University of Science and Technology, Shenzhen 518055, Guangdong, China

⁸ School of Materials Science and Engineering, Peking University, Beijing 100871, China.

⁹ School of Materials, Sun Yat-sen University Shenzhen Campus, Shenzhen, 518107, China.

¹⁰ School of Energy and Environment, City University of Hong Kong, Kowloon, Hong Kong S.A.R., China.

¹¹ School of Advanced Energy, Sun Yat-sen University Shenzhen Campus, Shenzhen, 518107, China.

[†] These authors contributed equally to this work.

* Corresponding authors: panfeng@pkusz.edu.cn; zhaoqinghe@fjnu.edu.cn; zhoud37@mail.sysu.edu.cn; shunnili@cityu.edu.hk.

Abstract

Although the anionic activity in layered oxide cathodes is partially understood, the mechanism by which it induces internal crack propagation under high-voltage conditions remains unclear, severely limiting stable cycle life and available capacity. In this study, a novel interior crack propagation mechanism is elucidated in LiCoO₂ (LCO) under harsh float conditions (60 °C, 4.6 V vs. Li/Li⁺). We reveal that with sequential phase transitions from O3 to O1 then to a cation-mixed (CM) phase during deep Li⁺ extraction process, significant lattice-mismatch stresses are generated to initiate internal cracks. Crucially, the accompanying anionic activity drives the migration of the

oxidized lattice O^{n-} ($0 < n < 2$) towards these cracks, where it even accumulates as molecular O_2 that can be released upon crack propagation to the surface. It is identified that the sustained O1-to-CM phase transition and the bulk lattice O^{n-} migration synergistically boost the interior crack propagation, ultimately leading to LCO disintegration and electrochemical degradation. This work not only deepens our understanding of the anionic activity in layered cathodes but also provides valuable guidance for the future design of stable LCO cathodes.

As a cornerstone cathode in commercial lithium-ion batteries (LIBs), layered $LiCoO_2$ (LCO) is extensively applied and dominates portable electronics.^{1, 2} However, to meet the escalating demand for higher energy density, operating LCO at elevated charge voltages of over 4.55 V (vs Li^+/Li) is crucial for accessing its full capacity.³ This high-voltage operation, however, inherently triggers severe structural instability and causes rapid electrochemical deterioration.³ Our previous investigations have illuminated that the high-voltage instability is intricately associated with the surface lattice curvature,⁴ irreversible phase transitions (e.g., the well-established $O3 \rightarrow H1-3 \rightarrow O1$ pathway),^{4, 5} and the activation and subsequent release of lattice O^{n-} upon deep delithiation,^{6, 7} which are coupled with cation disordering upon deep delithiation.⁸ These factors collectively contribute to the intrinsic degradation mechanisms of LCO.

Historically, research into LCO degradation at high voltages has predominantly focused on surface structural evolution, often describing a surface-to-bulk degradation pathway.^{9, 10, 11} This surface-centric view highlights two primary aspects. First, the gradual activation of surface lattice O^{n-} ($0 < n < 2$) leads to the migration and loss of O and Co,¹² which subsequently accelerates cation mixing while triggering irreversible phase transitions in the surface regions.¹³ As a result, this progressively transforms LCO surface to spinel and then rock-salt (RS) phases, critically impeding interfacial Li^+ transport, causing heterogeneous charge distribution,¹⁴ and ultimately leading to rapid capacity decay.^{15, 16} Second, electrolyte decomposition, particularly the formation of corrosive species such as HF, further exacerbates surface degradation and destabilizes surface-protective cathode/electrolyte interphase (CEI).^{9, 17, 18, 19} However, these surface-dominated mechanisms do not adequately explain the bulk structural degradation observed through crack formation. Despite their frequent observation and wide recognition, the precise mechanisms of crack propagation and their localized microscopic origins within LCO bulk remain significantly underexplored.

Upon deep delithiation, layered cathodes are known to undergo complex phase transitions, accompanied by significant anisotropic lattice distortions and accumulation of lattice-mismatch stress. This ultimately leads to the mechanical degradation and microcrack formation within individual particles.^{2, 8, 20, 21, 22} Generally, the high-voltage-induced O3-to-O1 phase transition, often proceeding through the H1-3 intermediate, is particularly pronounced, causing localized lattice dislocations and Co-O slab bending. These structural changes further induce drastic lattice variations with Co-O slab slippage, local structure collapse, and even oxygen reorganization.^{15, 23} In result,

substantial residual strain is accumulated with concentrated tensile and compressive stresses, initiating subsequent internal cracks and further particle fragmentation.²⁴ Recent studies have also highlighted a strong correlation between lattice strain-induced oxygen vacancy and reduced barriers for lattice oxygen release, which can intensify crack propagation along the (003) basal planes.²⁵ However, the precise mechanism by which the anionic activity contributes to the internal crack propagation in LCO upon deep delithiation is largely elusive.

Given the practical significance of developing high-voltage and high-energy-density LIBs, it is imperative to unravel the specific mechanisms and critical factors governing the interior crack propagation in LCO, a pervasive phenomenon and a major contributor to electrochemical degradation. To this end, we employed float tests,²⁶ a robust accelerated aging technique, to assess the long-term stability related to interior cracks. Herein, with various experimental techniques and theoretical calculations, we present the mechanistic study on the interior crack propagation within LCO. By carefully minimizing the influence of external factors (particularly the surface effects and electrolyte degradation), our results reveal a novel, synergistic phase transition and lattice O^{n-} migration-induced crack propagation mechanism. Specifically, upon nearly complete delithiation, the bulk lattice O^{n-} in O1 phase tends to migrate towards interior cracks and accumulates there, forming molecular O_2 trapped inside LCO bulk. This process is concurrently accompanied by the emergence of a cation-mixing (CM) phase. Crucially, subsequent float charge further exacerbates the internal stress along the O1/CM phase boundaries, significantly promoting the interior crack propagation. Our findings unequivocally demonstrate that the sustained O1-to-CM phase transitions and the migration of oxidized lattice O^{n-} species, coupled with molecular O_2 accumulation, synergistically drive the continuous propagation of interior cracks that eventually penetrate throughout the particle to surface, leading to gaseous O_2 release and severe LCO disintegration. With this fundamental mechanism underlying LCO crack propagation, our study may not only enhance the understanding of LCO degradation but also provide valuable guidance for the design of advanced layered oxide cathodes.

Phase transition of LCO upon float charge

A commercial LCO, characterized by a large D_{50} value of about 15 μm and containing trace Al (4500 ppm), Mg (1500 ppm), and Ti (500 ppm), was selected for this investigation into the degradation under float conditions. To assess the float-charge performance and induced degradation, coin-type LCO||Li cells were assembled and used. Float charge tests were performed by initially charging the cells to 4.6 V with a constant current of 100 mA g^{-1} (CC stage), followed by constant voltage charging at 4.6 V (CV stage) (Figure 1a). Fundamental characterizations, including scanning electron microscopy (SEM) (Figure S1), X-ray diffraction (XRD) coupled with Rietveld refinement (Figure S2), and transmission electron microscopy (TEM; Figure S3), confirmed a uniformly smooth surface and a pure α - NaFeO_2 -type layered structure with an $R\text{-}3m$ space group. However, the enlarged image in Figure S3b reveals the

125 presence of unavoidable inherent defects, such as curves in partial Co-O layers within
126 the commercial LCO, which is difficult to eliminate in actual production.

127 Under the high temperature float charge conditions, progressive crack formation was
128 distinctly observed during float charge from 0 to 40 h (**Figure S4**). Further investigation
129 into the temperature dependence of this degradation revealed a remarkable increase in
130 leakage current above 50 °C, underscoring the accelerated degradation kinetics at
131 elevated temperatures (**Figure S5**). Consistent with this, the extent of crack formation
132 varied with temperature: at 25 and 45 °C, only slight cracks appeared, whereas at 60 °C,
133 severe particle disintegration was evident (**Figure S6 and S7**). To further elucidate the
134 propagation of interior cracks, cross-sectional TEM analyses were conducted on the
135 samples after float charge at 25, 45, and 60 °C for an extended duration of 120 h (**Figure**
136 **S8**). These analyses demonstrated that while only minor interior cracks were observed
137 at 25 °C, extensive interior cracking became prevalent with increased temperature.

138 XRD analysis revealed that, after float charge for 120 h, O1 phase became dominant, a
139 notable deviation from the H1-3 phase typically expected at such a high voltage (**Figure**
140 **1b**). High-resolution scanning TEM (STEM) combined with geometric phase analysis
141 (GPA) revealed heterogeneous O1 domains embedded within the charged LCO bulk
142 before float charge (**Figure S9**). As an exceptionally compressed structure, O1 stacking
143 imposes pronounced stresses on the adjacent O3 matrix. This stress evidently intensifies
144 the curvature of the Co-O layers. To elucidate the correlation between stress evolution
145 and crack propagation under different float charging durations, systematic analyses
146 using 4D STEM were conducted (**Figures S10–S13**). Specifically, the stress
147 distribution mapping and line-scan profiles reveal a pronounced spatial correlation
148 between stress concentration regions and crack locations, suggesting that localized
149 stresses induced by phase transitions and other factors serve as a critical driving force
150 for crack advancement. To quantitatively assess the relationship between stress
151 concentration severity and crack geometry, a stress concentration factor K , defined as
152 $K = 1 + 2 \times (L/W)$, where L and W denote crack length and width, respectively, was
153 introduced (**Figure S14**). The results demonstrate a monotonic increase in K with
154 prolonged float charging time, rising from 3.37 at 5 h to 7.80 at 20 h, 9.50 at 40 h, and
155 ultimately reaching 20.26 at 120 h, indicating progressively intensified stress
156 concentration. More critically, a previously unreported phase characterized by a layer
157 spacing ($d = 2.087$ Å) approximately half that of the O1 phase ($d = 4.191$ Å) was
158 identified after float charge for 5 h (**Figure S15**). This newly formed phase, which we
159 term as the cation-mixed (CM) phase, originates from Co migration into the Li layer of
160 the O1 phase under certain stress disturbances (such as more curved Co-O layers and
161 more dislocations). Subsequently, driven by the internal stress, lattice oxygen
162 undergoes oxidation and migration. The oxygen vacancies generated during this
163 process, along with the induced local distortions and coordinatively unsaturated sites,
164 significantly lower the energy barrier for Co ions to hop into adjacent vacancies in the
165 Li layer. Ultimately, under the synergistic driving forces of internal stress, lattice
166 oxygen migration, and vacancies in the Li layer, a portion of Co ions irreversibly
167 migrate from the Co-O layer into the Li layer, thereby forming the structurally

disordered CM phase. Aberration-corrected STEM (AC-STEM) confirmed the atomic structure of the CM phase, suggesting that it forms a semi-coherent interface with the O1 phase (**Figure 1c**). The occurrence of this CM phase in LCO unambiguously indicates the increased cation disorder and significant lattice Co and O loss under severe float conditions, a key indicator of bulk degradation.²⁷

The emergence of CM phase is not merely a consequence of excessive Li⁺ extraction; rather, our findings strongly suggest that float charge actively facilitates the migration of bulk lattice Oⁿ⁻, which is considered the primary driver for the CM phase formation. This crucial mechanism was further confirmed by electron energy loss spectroscopy (EELS) performed across regions transitioning from the layered to CM phases (**Figure 1d**). In the EELS spectra, the pre-edge feature around 530 eV corresponds to the O 1s → O 2p–Co 3d hybridized states, while the main edge at around 541.5 eV is attributed to the core-level O 1s → O 2p transitions.²⁸ The intensity ratio of the pre-edge (I_p) to the main edge (I_m), denoted as I_p/I_m , directly correlates with the concentration of oxygen vacancies (Ov).²⁹ Remarkably, as shown in **Figure 1e**, the Ov concentration was found to decrease upon transition from the layered to the CM phase. This indicates that the substantial oxygen loss from the layered phase drives the generation of under-coordinated Co ions, which then migrate into the Li layers to form the structurally rearranged and potentially more stable CM phase. Concurrently, the Co L_3 -edge peak shifts to lower energy in the CM phase, further corroborating a reduction in Co oxidation state (**Figure 1f**), a direct consequence of O loss and structural rearrangement within LCO lattice.

Formation of molecular O₂ trapped in interior cracks

As discussed previously, the formation of CM phase within LCO induces significant internal stresses at its boundary with the layered phase due to their distinct lattice configurations, which serve as the critical initiation sites for interior cracks. To investigate the origin and propagation of these cracks, particularly those formed after float charge for 5 h, high-resolution STEM coupled with GPA was performed (**Figure 2a-c**). Analysis reveals concentrated regions of intense tensile (shown by red) and compressive (shown by green) stress localized at the crack tips, indicating these stresses as the primary driving force for crack propagation (**Figure S9**). Further structural analysis of the crack-tip region shows the coexistence of layered O1 and CM phases (**Figure S15**), where the inherent lattice mismatch at their interface acts as the starting point for these internal cracks. Continued float charge under high-voltage conditions further promotes this crack propagation, ultimately leading to severe particle disintegration.

Beyond stress-induced crack initiation, our comprehensive analyses, including EELS, electron paramagnetic resonance (EPR), and *in situ* differential electrochemical mass spectrometry (DEMS), also elucidate the crucial role of bulk lattice Oⁿ⁻ migration in the crack propagation mechanism. Specifically, the O K -edge EELS spectra reveal a significant increase in Ov concentration from LCO bulk towards the crack tips (**Figure**

210 **2e,f**), indicating the substantial loss of bulk lattice O^{n-} at these defect sites. This
 211 observation supports our hypothesis that the interior cracks, serving as the *in situ*-
 212 formed defects, provide preferential and shorter diffusion pathways for the O^{n-}
 213 migration and, critically, the confined spaces for molecular O_2 accumulation. EPR
 214 measurements directly evidence the presence of molecular O_2 within these interior
 215 cracks (**Figure 2g**): characteristic resonance signals between 5500 and 9000 G were
 216 detected in the LCO after float charge for 10 h.³⁰ However, the LCO that undergoes
 217 float charging at 25 °C does not produce a significant O_2 signal.(**Figure S16**) The
 218 release of this trapped molecular O_2 was further probed by *in situ* DEMS (**Figure 2h**
 219 and **S17**). **Figure S18** displays the resonant inelastic X-ray scattering (RIXS) spectra
 220 comparing pristine LCO with the LCO after float charging at 60 °C and 4.6 V for 10 h
 221 and 20 h. The absence of detectable signals in pristine LCO, in stark contrast to the
 222 clear RIXS signatures observed for the LCO after 10 h and 20 h of charging,
 223 substantiates the formation of O_2 in the LCO bulk. Our findings suggest that the
 224 accumulated molecular O_2 within the interior cracks induces additional internal stresses
 225 that drive crack propagation, in addition to the lattice-mismatch induced internal
 226 stresses. Besides, DEMS reveals pronounced O_2 evolution between 8 and 20 h of float
 227 charge, which coincides with an increase in the float charge capacity (C_{float}) from 10 to
 228 20 h and indicates a close correlation between gaseous O_2 release and C_{float} increase.
 229 This underlying correlation can be elucidated as follows. First, as float charge proceeds,
 230 the concentration of oxidized lattice O^{n-} increases, and these O^{n-} ions migrate into the
 231 interior cracks and form molecular O_2 there. When these interior cracks propagate
 232 sufficiently to reach the particle surface, the trapped O_2 is released, producing the
 233 observed surge in O_2 signal. Second, the subsequent propagation of these cracks to the
 234 surface exposes fresh and highly active LCO interfaces (with active species like
 235 Co^{4+}/O^{n-}) to the electrolyte, which triggers intense interfacial side reactions. These
 236 reactions are directly responsible for the observed increase in C_{float} . This precise
 237 correlation between O_2 evolution and current response supports the proposed
 238 mechanism of HT floating-induced failure, which originates from bulk oxygen
 239 oxidation and crack propagation.

241 **Propagation of interior cracks**

242 To gain a deeper understanding of the dynamic interior crack propagation, transmission
 243 X-ray microscopy (TXM) coupled with X-ray absorption near-edge structure (XANES)
 244 and extended X-ray absorption fine structure (EXAFS) was further employed (**Figure**
 245 **3a,b**). In TXM images, lighter regions correspond to the lower-valence Co, whereas
 246 darker regions indicate the higher-valence Co. After float charge for 5 h, LCO bulk
 247 exhibits a uniform distribution of Co valence, suggesting that the overall bulk structure
 248 remains largely intact at this early stage. However, a stark contrast emerges after float
 249 charge for 40 h, i.e., the Co valence distribution becomes highly heterogeneous,
 250 indicating extensive bulk degradation that is attributed to the large-scale migration of
 251 lattice O^{n-} and the propagation of interior cracks. Complementary TEM images further
 252 corroborate these findings, revealing a pronounced CM phase and intensified crack

growth between 5 and 40 h of float charge (Figure S19). This highlights the close correlation between the O1-to-CM phase transition and the propagation of interior cracks. Cross-sectional TEM analyses of the LCOs after float charge for 5, 20, and 40 h further show that the interior cracks preferentially initiate along the (003) planes and subsequently coalesce into larger and more destructive fissures during prolonged float charge (Figure S20). Furthermore, TXM tomography provides a powerful three-dimensional visualization of the internal crack networks (Figure 3c). Its quantitative analysis shows that the volume fraction of interior cracks dramatically increases from 2.06% to 5.51% between 5 and 40 h of float charge, clearly illustrating the rapid and extensive propagation of these interior cracks.

XANES was also employed to monitor the dynamic changes in lattice O^{n-} migration and local structural evolution (Figure 3d and S21a). XANES analysis of the Co absorption edge reveals a shift to higher energy after 20 h of float charge, consistent with an increase in Co oxidation state associated with the O3 $\rightarrow$ O1 phase transition.³¹ However, it moves to the opposite direction after further float charge for 120 h, reflecting a decrease in Co valence due to the substantial lattice O^{n-} release, a key event in the degradation process. Further insights into the local structural evolution were enabled by EXAFS analysis (Figure 3e and S21b). The nearest Co–O coordination shell first shifts to a reduced radial distance after float charge for 20 h, signifying shorter Co–O bond lengths characteristic of the O1 phase.³² Whereas, it moves back after 120 h, indicating bond elongation, consistent with the formation of the CM phase. Unlike Co–O, the first Co–Co coordination shell exhibits a sustained decrease in radial distance throughout the float charge process, directly attributed to the transition of O1 to CM phase. Moreover, the diminished peak intensities and contracted distances at higher R -values (4–10 Å) for other Co–O and Co–Co coordination shells after float charge for 120 h unequivocally reveal the widespread existence of structural disorder within LCO.

Theoretical calculations on the migration of bulk lattice O^{n-}

These experimental observations above, including EELS, *in situ* DEMS, XANES, and EXAFS, clearly illustrate the migration of bulk lattice O^{n-} to the interior cracks in LCO, which is at the core of the proposed mechanism linking molecular O_2 accumulation and crack propagation. To further elucidate the precise underlying atomic-scale mechanisms, theoretical investigations were conducted. Here, we employed Monte Carlo (MC) simulations and percolation theory to evaluate the capability of O^{n-} migration within LCO.³³ Our theoretical model was predicated on the understanding that during the O1-to-CM phase transition, Co migrates from its original CoO_6 octahedron into the interlayer region, leaving behind several types of under-coordinated O ions (i.e., connected with fewer than three Co ions) that tend to be oxidized to O^{n-} . Crucially, these redox-active O^{n-} species can form a continuous migration pathway within the anion sublattice, similar to the case of Li-rich layered cathodes. For computational efficiency, we simplified the anion sublattice into a two-dimensional cellular network representing O^{n-} migration from the bulk to the boundary (Figure 4a).

In these MC simulations, the redox-active lattice O^{n-} species are defined as the under-coordinated O ions induced by Co migration. A migration pathway for lattice O^{n-} is established when two adjacent redox-active O^{n-} sites are connected to the same Co ion, and once the concentration of these redox-active lattice O^{n-} sites reaches a critical threshold, a continuous network for O^{n-} migration emerges.

As the O1-to-CM phase transition progresses, the proportion (denoted as α) of Co migrating to the interlayer region increases. The simulations demonstrate that at the early stages of Co migration ($\alpha < 0.2$), the connection probability (p) of redox-active O^{n-} sites—representing the likelihood of forming a continuous pathway from one boundary to another—remains less than 0.05 (**Figure 4b**). However, as α surpasses 0.5, p increases sharply to over 0.9, indicating that nearly the entire network of redox-active O^{n-} sites becomes fully connected to the boundary. This is known as the percolation of redox-active O^{n-} in the anion sublattice, upon which the oxidized lattice O^{n-} can successfully migrate over long distances to the LCO surface or cracks, ultimately leading to the release of gaseous O_2 . Significantly, our analysis shows that the distribution of Co ions within the CM phase exactly aligns with the case of $\alpha = 0.5$; that is, half Co ions are located in their original layer of corresponding O1 phase, while the other half occupy the interlayer region. This suggests that before the complete formation of the CM phase, a considerable fraction of redox-active lattice O^{n-} can diffuse to the boundary during the Co migration, thus generating a substantial amount of O_v in the lattice. When α approaches 0.5, the accumulated O_v triggers an atomic rearrangement of the anion sublattice, culminating in the formation of the CM phase, which, as observed experimentally, exhibits a much lower O_v concentration. These results suggest that the formation of CM phase and the long-range migration of lattice O^{n-} are mutually synchronized, and this theoretical insight explains our experimental observations that the accumulation of molecular O_2 at the interior cracks is consistently accompanied by the CM phase emergence near the crack tips.

Beyond above migration pathways, we further studied the atomic-level mechanisms facilitating these transformations. The formation of O dimers within LCO bulk is considered a prerequisite for generating molecular O_2 , yet its mechanistic role in the O1-to-CM phase transition remained unclear. As shown in **Figure 4c**, we investigated the scenario where one Co ion migrates near an O dimer formed within corresponding CoO_6 octahedron in the initial configuration. Both the intralayer and interlayer migrations of Co ions were evaluated. Our calculations demonstrate that the presence of an O dimer significantly reduces the energy barriers by 0.65 and 0.68 eV for intralayer and interlayer Co migration, respectively (**Figure 4d**). Even in the presence of some residual Li (with SoC = 90.9%, **Figure S22**), the formation of O dimer can still accelerate the Co migration. This finding strongly indicates that the formation of O dimers actively accelerates the Co migration, thereby facilitating the O1-to-CM phase transition. Given the low kinetic barriers in the delithiated O1 phase, the Co migration likely precedes both the long-range migration of lattice O^{n-} and the propagation of interior cracks, which is consistent with the experimentally observed rapid formation of the CM phase within LCO bulk.

Degradation mechanism

In addition to bulk degradation, we also observed severe surface degradation of LCO under float charge conditions. This pronounced degradation is primarily attributed to two factors. First, an elevated state of charge (SOC) at the LCO surface promotes the generation of highly oxidative Co^{4+} and O^{n-} species. Second, this environment accelerates the electrolyte decomposition, leading to the formation of corrosive species such as HF and HPO_2F_2 . Nuclear magnetic resonance (NMR) analysis of the electrolytes cycled at 25, 45, and 60 °C demonstrates the temperature-dependent increase in the characteristic peaks of PO_2F_2^- , a byproduct of HF, at chemical shifts of -83.0 and 85.5 ppm (**Figure S23**). This indicates enhanced formation of HF and HPO_2F_2 and severe SiO_2 -based glass corrosion at elevated temperatures. Correspondingly, X-ray photoelectron spectroscopy (XPS) signal of Co on Li anodes from the LCO||Li cells after float charge for 120 h intensifies markedly with increased temperature, directly reflecting the accelerated HF-induced Co dissolution from the LCO surface (**Figure S24**).

At 60 °C, LCO surface undergoes significant morphological and structural changes. A highly porous surface layer, exceeding 100 nm in thickness, is observed (**Figure S25**), predominantly as a consequence of the severe Co and O loss. Selected-area electron diffraction (SAED) from this region confirms the coexistence of O3, O1, spinel, and rock-salt (RS) phases, alongside a distorted lattice, indicative of numerous vacancies and crystal-plane slippage. Besides, cross-sectional STEM further reveals a ~5 nm thick amorphous surface film, underlain by a 5-10 nm RS subsurface layer and a mixed O1/CM/spinel bulk phase (**Figure S26**). Meanwhile, the CEI exhibits a thick and porous morphology (**Figure S27**), reflecting aggressive interfacial side reactions. Collectively, these integrated observations demonstrate that float charge at 60 °C induces pronounced surface degradation driven by intensified interface reactions, ultimately leading to extensive Co/O depletion and porous layer formation.

Figure 5 comprehensively illustrates the proposed mechanism of interior crack propagation in LCO, highlighting the synergistic interplay between sustained $\text{O3} \rightarrow \text{O1} \rightarrow \text{CM}$ phase transitions and the collective lattice O^{n-} migration towards the interior cracks. During float charge, continuous Li^+ extraction from the LCO bulk is accompanied by the emergence of uneven O1 domains, which evolve into the CM phase under exacerbated disturbance such as stresses from Co-O lattice curves. The CM phase transition is also promoted by oxygen oxidation and migration. Oxidized lattice O^{n-} species migrate into Li-layer sites, which significantly lowers the energy barrier for Co migration, thereby driving the irreversible formation of the CM phase. These transitions generate more pronounced internal stresses due to lattice mismatch, contributing to the generation and propagation of the interior cracks along the O1/CM boundaries. Simultaneously, as elucidated in previous sections, lattice O^{n-} migrates towards and accumulates within these interior cracks as molecular O_2 . The trapped O_2 generates additional localized gas pressure at the crack tip, which synergizes with the lattice

mismatch-induced stress, creating a combined driving force that effectively wedges open and accelerates crack propagation. Upon propagation to the particle surface, the accumulated O₂ is released, and the newly exposed and highly active interfaces undergo aggressive side reactions with the electrolytes (**Figure S28**). Thus, the synergy between sustained phase transitions and Oⁿ⁻ migration continuously drives crack propagation from the bulk to the surface, culminating in severe LCO disintegration and electrochemical degradation.

As the desire for increased energy density of LCO cathodes becomes increasingly urgent in industry, there is a pressing requirement to develop LCO with high floating durability at 4.6 V and 60 °C. Our above results have comprehensively demonstrated the interior crack propagation mechanism of LCO, including the sustained O1-to-CM phase transition, accumulation of molecular O₂ in interior cracks, the propagation of cracks in the bulk region of LCO, and severe surface Co/O loss induced by side reactions. Based on these findings, we propose the following strategies that can effectively enhance the floating durability. Firstly, suppressing the bulk O1-to-CM phase transition is essential, as it serves as the origin for long-range migration of lattice Oⁿ⁻ and subsequent propagation of interior cracks. Secondly, it is crucial to develop advanced electrolytes that remain stable at high temperatures (such as 60 °C), thereby reducing electrolyte decomposition and the formation of corrosive acidic species (e.g., HF), which in turn mitigates surface Co/O loss and interfacial side reactions.

And from the perspective of bulk durability, structural modifications, especially doping by heteroatoms like Al, have been validated as feasible approaches to enhance the stability of LCO lattice, while their effects in alleviating both the extensive migration of lattice Oⁿ⁻ and the propagation of the interior cracks remain insufficiently explored. To this end, the modified LCO is synthesized via incorporating 8000 ppm Al in LCO bulk (Al-reinforced LCO), with the detailed synthesis provided in SI. Specifically, due to the strong covalent Al-O bonds, the bulk lattice Oⁿ⁻ is more effectively stabilized within the crystal structure, thereby significantly enhancing the bulk stability during float testing. **Figure S29** compares the XRD patterns of the Al-reinforced LCO after floating for 120 h. At 25 °C, the H1-3 phase predominates the bulk structure of Al-reinforced LCO, with some O3 phase retained. Even at 60 °C, the bulk structure is still dominated by the H1-3 phase, with only a small amount of O1 phase detected, which contrasts with the bare LCO (**Figure 1b**). This indicates that the heavy Al doping can effectively suppress the O1-to-CM phase transition, thereby fundamentally reducing the bulk internal stresses and resulting in the more integrated LCO with fewer cracks (**Figure S30**). Consequently, comparing to the commercialized LCO, the Al-reinforced LCO exhibits the much lower leakage current without obvious gaseous O₂ release upon floating (**Figure S31 and S32**).

Conclusion

In this work, we provide a comprehensive elucidation of the intricate mechanism governing interior crack propagation in LCO during float charge at 4.6 V with elevated

temperatures. Upon float charge, the substantial Li^+ extraction triggers sequential $\text{O3} \rightarrow \text{O1} \rightarrow \text{CM}$ phase transition. The critical irreversible O1-to-CM phase transition plays a dual role in LCO degradation: it not only generates significant internal stresses that concentrate near the interior crack tips, but also actively promotes the long-range migration of bulk lattice O^{n-} , which in turn leads to the accumulation of molecular O_2 within these interior cracks. The synergistic interplay among these factors, including the intensified internal stresses from lattice mismatch at the O1/CM interface and the accumulated molecular O_2 within interior cracks, serves as the primary driving force for crack propagation. As these cracks propagate to the surface, the trapped molecular O_2 is released as gas. Such cascading processes ultimately lead to severe LCO disintegration with pronounced Co and O depletion, further exacerbated by aggressive surface side reactions occurring along the newly exposed crack surfaces. Overall, our study highlights the critical and interconnected roles of the O1-to-CM phase transition and the long-range lattice O^{n-} migration in the interior crack propagation of LCO, providing invaluable mechanistic insights into the rational design for advanced layered cathodes like LCO with superior bulk stability and long-term durability.

Methods

Material characterization. The commercialized LCO used in this work was purchased from Pulead Technology Industry Co., Ltd. The XRD was performed on a Bruker D8 Advance diffractometer with $\text{Cu K}\alpha$ radiation ($\lambda = 0.154 \text{ nm}$), and Rietveld refinements of XRD patterns were carried out using the FULLPROF software. Sample morphology and elemental distribution were explored by SEM (Zeiss SUPRA-55) with energy-dispersive X-ray spectroscopy (EDS, Oxford X-Max 80). The CEI was characterized by cryogenic TEM (cryo-TEM): charged electrode powders were scraped in an argon glove box, dispersed onto copper grids, mounted on a Fischione 2550 cryo-transfer holder, and transferred to a field-emission TEM (JEOL JEM-3200FS). High-resolution STEM images and EELS analysis were performed by the double aberration-corrected FEI Titan Themis G2 equipped with a Gatan GIF Quantum ERS spectrometer. Lattice strain fields were analyzed using GPA based on high-resolution STEM images. TEM samples were prepared using the focused ion beam microscope (FIB, Thermo Fisher SCIOS). *In situ* DEMS measurements employed an EL-CELL EC-DEMS cell and a Hiden HPR-20 EGA spectrometer. XPS was carried out on a Thermo Fisher ESCALAB 250Xi using a monochromic $\text{Al K}\alpha$ X-ray source. The binding energy scale was calibrated by the C 1s peak at 284.8 eV. XANES and EXAFS spectra were collected on an easyXAFS300⁺ spectrometer. TXM tomography data were acquired at the 3D nano imaging beamline (BL18B) of the Shanghai Synchrotron Radiation Facility (SSRF). ^{19}F NMR spectra were recorded on a Quantum-Plus 400 MHz spectrometer at 298 K. EPR spectra were collected on a Bruker EMX plus 10/12 spectrometer equipped with a dual-mode microwave cavity and an Oxford ESR910 liquid helium cryostat. The perpendicular-mode EPR spectra were acquired at a microwave frequency of 9.645 GHz under 60 K using 2 mW microwave power and 2 G modulation amplitude. The 4D STEM experiments were conducted on a JEOL JEM-ARM300F2 transmission

electron microscope operated at 300 kV, equipped with a probe-forming aperture of 1.3 mrad convergence semi-angle. A Medipix3RX MerlinEM direct electron pixelated detector was employed to capture the 4D STEM dataset. Strain mapping was performed using Python code with reference to the py4DSTEM project.³⁴ RIXS experiments were conducted at beamline U41-PEAXIS at BESSY II, located at Helmholtz-Zentrum Berlin (HZB). A vacuum suitcase was used to transfer the sample from an N₂-filled glovebox to the test chamber. The spectrometer was positioned at specular conditions relative to 60° scattering angle and was optimized to a combined resolution of 90 meV using a carbon tape. The O K-edge RIXS spectra of the samples were collected at an excitation energy of 531.0 eV. The acquisition time for each pattern was 30 min.³⁵

Electrode preparation and electrochemical methods. The cathode electrodes were fabricated by mixing LCO (90 wt%), polyvinylidene fluoride (PVDF, 5 wt%), and acetylene black (5 wt%) in N-methyl-2-pyrrolidone (NMP) solvent, followed by casting the slurry onto an aluminum foil and dried at 120 °C in vacuum for 12 h. The mass loadings of active materials in the cathode films were ~6 mg cm⁻². The electrochemical performance was assessed by coin-type (CR2032) half cells using Li as the anode, Celgard 2400 polypropylene (PP) membrane as the separator, 1.0 M LiPF₆ in ethyl carbonate (EC) / ethyl methyl carbonate (EMC) / fluoroethylene carbonate (FEC) with a volume ratio of 1:1:1 as the electrolyte. The float charge measurements were carried out using a NEWARE battery test system at 25, 45, 50, 55, and 60 °C. In the float charge procedure, the LCO||Li cells were firstly charged to 4.6 V upon a constant current (0.5 C), and then the cells were continuously charged upon a constant 4.6 V voltage for 120 h.

Theoretical calculation methods. Crystal structure simulation was performed based on density functional theory (DFT) in the Vienna *ab initio* simulation package (VASP 5.4.4). The projector-augmented wave (PAW) approach was used to represent the ion/electron interaction. The generalized gradient approximation with the Perdew, Burke, and Ernzerhof (PBE) method was used to process the electronic exchange-correlation function. In the case of weaker interactions, using GGA can adversely impact the calculation accuracy. The cut-off energy was set to 500 eV with an energy precision of 10⁻⁶ eV, and the atomic positions were fully relaxed until the maximum force on each atom was less than 10⁻⁴ eV Å⁻¹. In all the calculations, the Brillouin zone was sampled by a Monkhorst-Pack k-point grid with a density of at least 0.2 Å⁻¹. In order to correctly characterize the localization of transition-metal (TM) *d*-electrons, the GGA+U method (Dudarev's approach³⁶) was used to account for the strong correlation interactions, with an effective U parameter (U_{eff}) of 5.0 eV applied to the Co 3d orbitals. DFT-D3 semi-empirical van der Waals correction was applied to deal with the dispersion force during structural relaxation. The migration barriers of ions were calculated using the method of climbing-image nudged elastic band (CINEB). MC simulations and traversal scanning techniques were used to study the migration of O²⁻ ions with redox activity. The penetration probability of O²⁻ ions was correlated with the

occupancy of Co in the TM layer. Due to the difficulty of large-scale migration of O^{n-} ions between different Co-O layers, when simulating the migration path of O^{n-} ions, the system was simplified as a two-dimensional cellular network of 100×100 , 200×200 , and 300×300 simulated based on each honeycomb cell as the basic unit. In addition, it is specified that if the O^{n-} ion is connected to less than three Co ions, it is considered to have redox activity. The path of O^{n-} ion migration is formed between two nearest redox active O sites, and both sites need to be connected to the same Co ion site in order to be considered as a continuous path. The nearest neighbor path between the O atoms around the vacancy is considered to be connected. When the redox-active O site network is connected to a periodic boundary, it is considered permeable. In the MC simulation and traversal scanning technology, the process begins with the initial configurations where Co atoms in the Co-O layer are fully filled. As the phase transition proceeds, the random transformation of Co atoms into vacancies is used to represent the cross-layer migration of Co, with α of Co being set until a specified ratio (ranging from 0.1 to 0.9) is achieved. Subsequently, reactive O^{n-} ions are determined based on the specified conditions, and traversal scanning technology is employed to calculate the proportion of reactive O^{n-} ions at the successful infiltration boundary relative to all reactive O^{n-} ions, thereby obtaining the p . Finally, vacancy positions for Co are randomly generated 10,000 times, and the p values obtained from each MC scan are averaged to enhance the reliability of the results.

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Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to W. Z., Z. L., M. L., S. L., D. Z., Q. Z. and F. P.

Author contributions

F. P., Q. Z., D. Z., and S. L. conceived the work and designed the experiments. W. Z., Z. L., and M. L. carried out the electrochemical measurements. W. Z. and Z. L. carried out the SEM, FIB, TEM, and XRD characterizations, and M. L. H. Y., H. R., and X. W. assisted with the data analysis. H. R. and X. W. performed the DEMS measurement. W. Z., Z. L., and G. C. carried out the EPR, XANES, EXAFS, and TXM characterizations, M. Z., M.W., Y.S. and X. Z. carried out the 4D STEM characterizations, and M. L. S. W., J. W., and B. F. assisted with the data analysis. M. L., Z. C., and W. C. carried out the theoretical calculation, and W. Z., Z. L., C. L., and S. L. assisted with the result analysis. W. Z., Z. L., and M. L. wrote the manuscript, and F. P., Q. Z., D. Z., and S. L. assisted with the manuscript revision. All authors agreed and edited the manuscript.

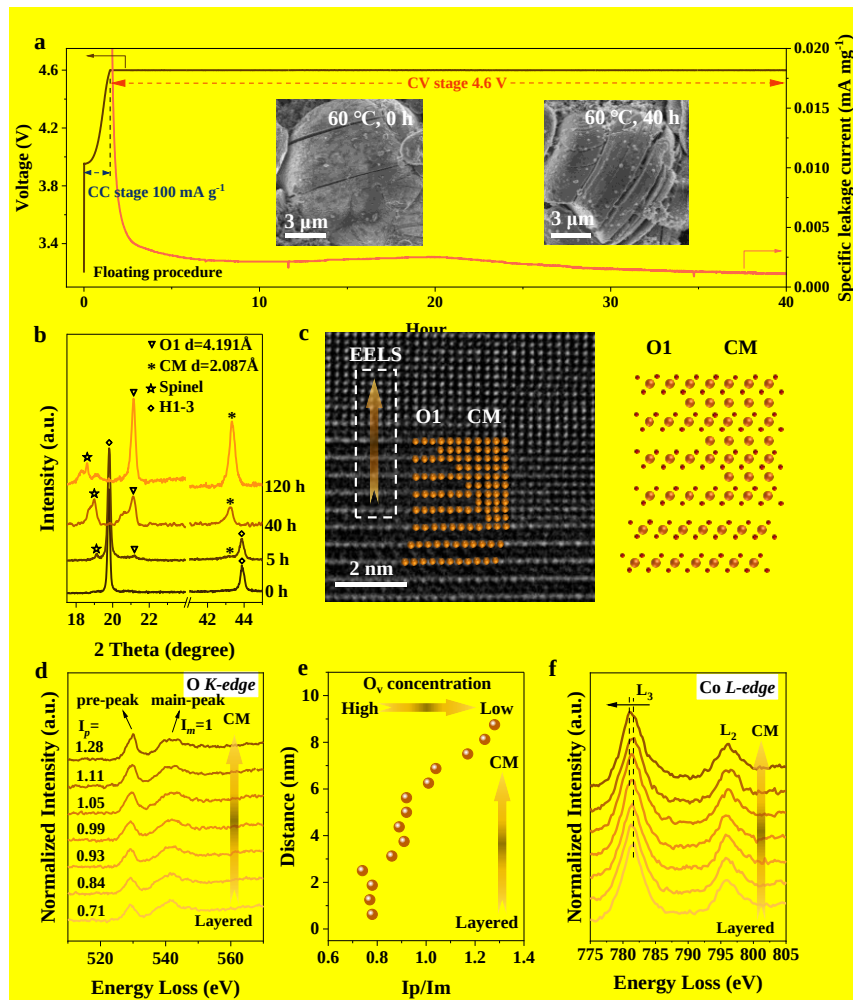


Figure 1. Structural evolution during float charge

(a) Electrochemical charge profile of the LCO||Li cell, with the insets showing the particle morphology after 5 and 40 h of CV stage.

(b) *Ex situ* XRD patterns of LCO after float charge at 60 °C for different durations.

(c) STEM image of LCO after float charge for 5 h at 60 °C (left) and corresponding structural sequencing models of O1 and CM phases (right), with the yellow arrow indicating the EELS line-scan direction.

(d) O K-edge EELS spectra collected along the path in (c), with I_p/I_m labeled for each spectrum.

(e) Distribution of O_v derived from (d).

(f) Co L-edge EELS spectra collected along the path in (c).

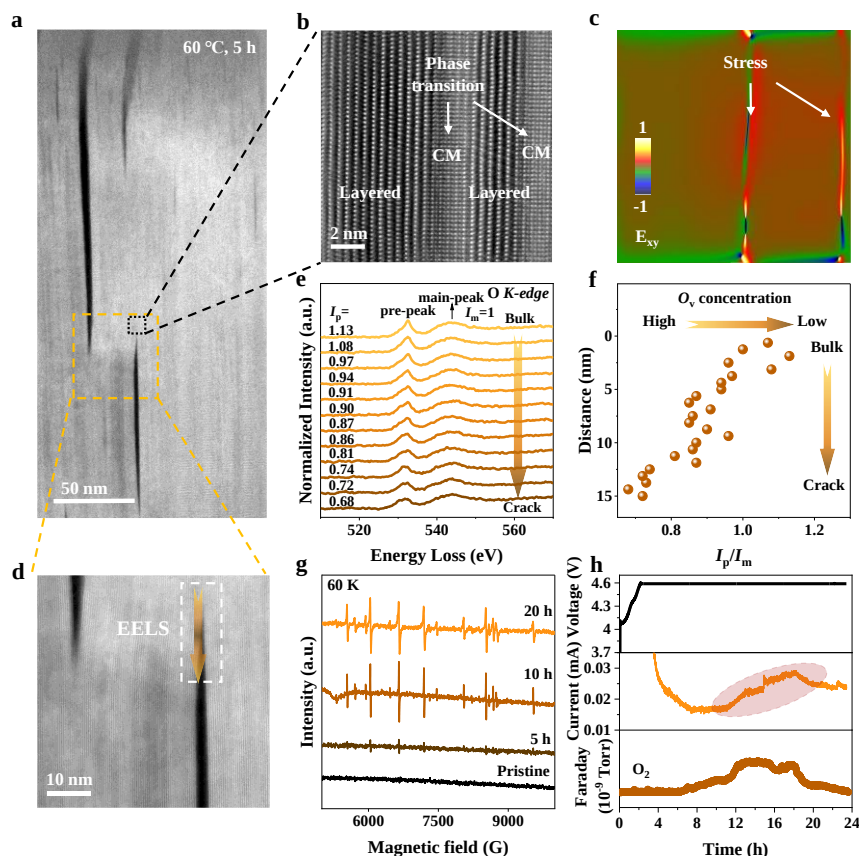


Figure 2. Evolution of interior cracks and lattice oxygen during float charge

(a) Cross-sectional STEM image after float charge for 5 h at 60 °C.

(b) High-resolution STEM image of corresponding region at the crack tip in (a).

(c) GPA map of corresponding region in (b).

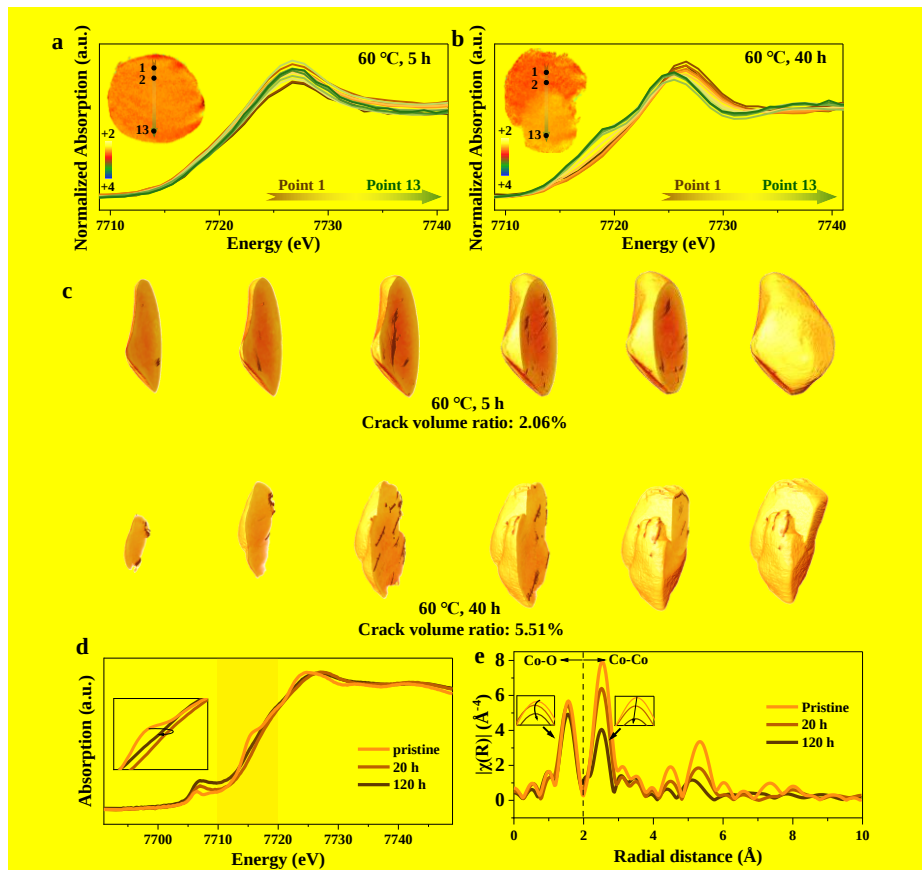
(d) Enlarged STEM image of corresponding region in (a), with the white box and yellow arrow indicating the respective region and direction of the O K-edge EELS spectra.

(e) O K-edge EELS spectra collected along the path in (d).

(f) Distribution of Ov derived from (e).

(g) EPR spectra of LCO after float charge for various durations at 60 K.

(h) *In situ* DEMS profile, with voltage profile (top), leakage current (middle), and corresponding O₂ signal (bottom).



697 **Figure 3. Evolution of Co valence and internal cracks during float charge**

698 (a, b) Co K-edge XANES spectra of LCO after float charge for 5h (a) and 40 h (b) at
699 60 °C, with Co valence distribution from XANES shown as the insets.

700 (c) Three-dimensional visualization of the internal crack networks after float charge for
701 5 (top) and 40 h (bottom) at 60 °C.

702 (d) Co K-edge XANES spectra of LCO after float charge for different durations at 60 °C,
703 with the inset showing a magnified view at around the half-maximum of the absorption
704 edge.

705 (e) Corresponding Fourier-transformed k3-weighted EXAFS spectra.

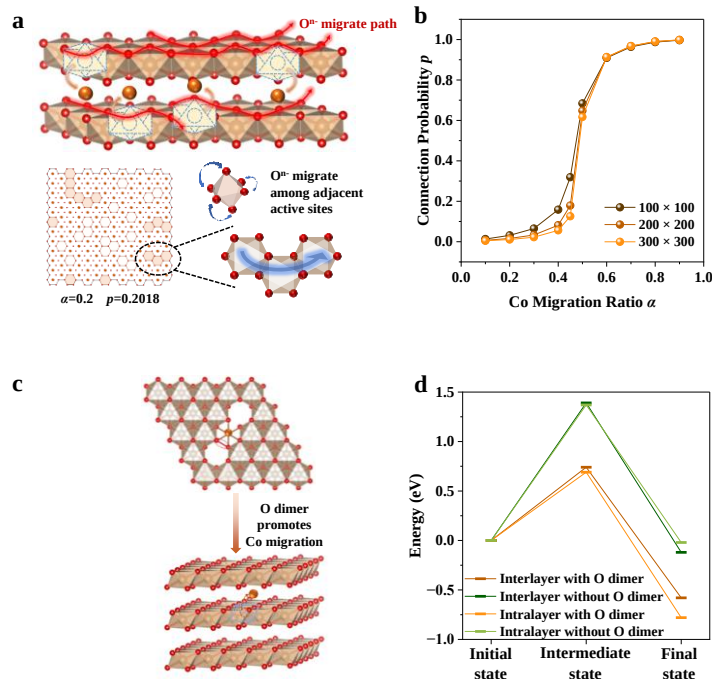


Figure 4. Theoretical investigation of the migration of lattice O^{n-} and Co during the O1-to-CM phase transition

(a) Migration pathway of the redox-active lattice O^{n-} after Co migration to the interlayer region.

(b) Relationship between Co migration ratio (α) and the connection probability of O^{n-} (p).

(c) Schematic illustration of O dimer facilitating Co migration.

(d) Migration barriers for Co in the presence and absence of bulk O dimer.

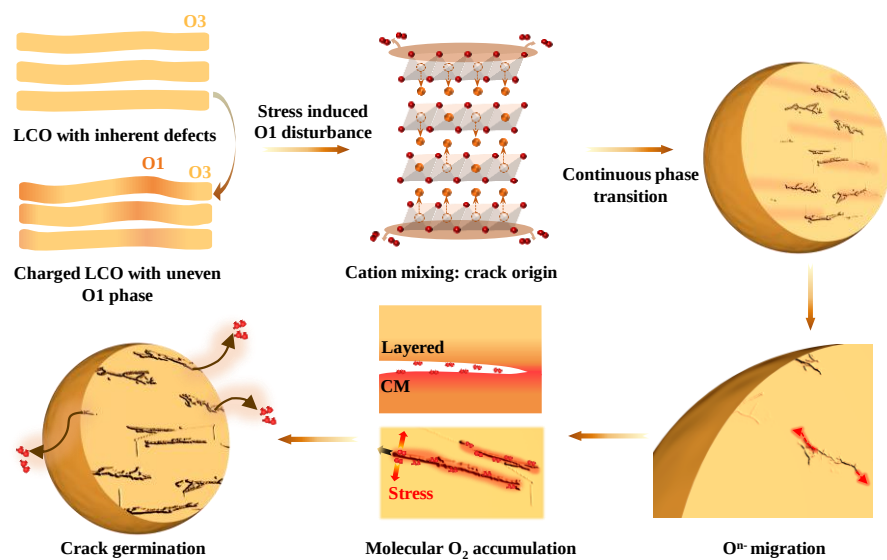


Figure 5. Schematic interior crack propagation under float charge